

WTPR5

Problem Statement

Implement the sample program demonstrating the use of Servlet

Folder structure

D:\apache-tomcat-9.0.85\webapps\STUDENTAPP\

```
├── index.html           ← HTML form for student registration
├── success.jsp         ← Simple confirmation page
├── WEB-INF/
│   ├── web.xml         ← Servlet configuration
│   ├── classes/
│   │   └── com/
│   │       └── example/
│   │           └── StudentServlet.class ← Compiled servlet file
│   └── lib/
│       └── mysql-connector-j-8.0.33.jar ← JDBC driver for MySQL
```

web.xml Configuration File

```
<web-app xmlns="http://java.sun.com/xml/ns/javaee" version="3.0">
  <servlet>
    <servlet-name>StudentServlet</servlet-name>
    <servlet-class>com.example.StudentServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>StudentServlet</servlet-name>
    <url-pattern>/register</url-pattern>
  </servlet-mapping>
</web-app>
```

MySQL Database (Table Structure)

```
CREATE TABLE students (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100),
  course VARCHAR(100)
);
```

The screenshot shows the MySQL Workbench interface for the 'students' table in the 'studentdb' database. The 'Table structure' tab is active, displaying the following columns:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	name	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	email	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	course	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Below the table structure, there are options to 'Check all', 'With selected', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Fulltext'. There are also buttons for 'Print', 'Propose table structure', 'Move columns', and 'Normalise'. At the bottom, there is an 'Indexes' section showing the 'PRIMARY' index on the 'id' column.

HTML Form Submission (Form Page) index.html

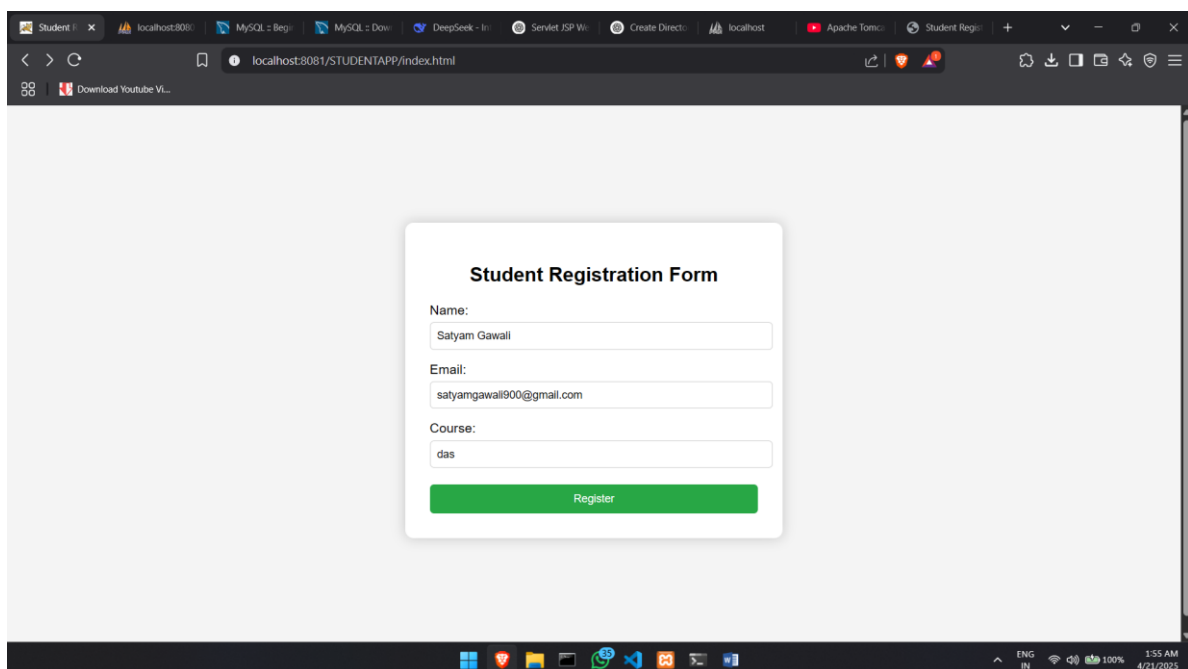
```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration</title>
  <link rel="stylesheet" type="text/css" href="style.css">
</head>
<body>
  <div class="form-container">
    <h2>Student Registration Form</h2>
    <form action="register" method="post">
      <label>Name:</label>
      <input type="text" name="name" required>
      <label>Email:</label>
      <input type="email" name="email" required>
      <label>Course:</label>
      <input type="text" name="course" required>
      <input type="submit" value="Register">
    </form>
  </div>
</body>
</html>
```

Style.css

```
body {
  font-family: Arial, sans-serif;
  background: #f4f4f4;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}
.form-container {
  background: white;
  padding: 30px;
  border-radius: 10px;
  box-shadow: 0 0 15px rgba(0,0,0,0.2);
  width: 400px;
}
h2 {
  text-align: center;
}
label {
  display: block;
  margin-top: 15px;
}

input[type="text"],
input[type="email"] {
  width: 100%;
  padding: 8px;
  margin-top: 5px;
  border: 1px solid #ccc;
  border-radius: 5px;
}

input[type="submit"] {
  margin-top: 20px;
  width: 100%;
  padding: 10px;
  background: #28a745;
  color: white;
  border: none;
  border-radius: 5px;
  cursor: pointer;
}
```

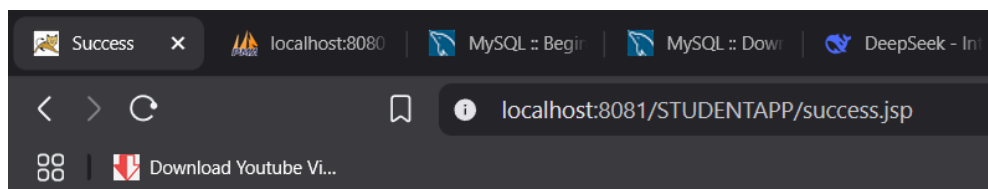


StudentServlet.class

```
package com.example;
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/register")
public class StudentServlet extends HttpServlet {
    public StudentServlet() {
    }
    protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws
ServletException, IOException {
        String var3 = var1.getParameter("name");
        String var4 = var1.getParameter("email");
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection var5 =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "");
            String var6 = "INSERT INTO students (name, email) VALUES (?, ?)";
            PreparedStatement var7 = var5.prepareStatement(var6);
            var7.setString(1, var3);
            var7.setString(2, var4);
            var7.executeUpdate();
            var5.close();
            var2.sendRedirect("success.jsp");
        } catch (Exception var8) {
            var8.printStackTrace();
            var2.getWriter().println("Database Error: " + var8.getMessage());
        }
    }
}
```

Database Connection (Success or Error Message) success.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head>
    <title>Success</title>
</head>
<body>
    <h2>🎉 Registration Successful!</h2>
    <p>Thank you for registering.</p>
</body>
</html>
```



 **Registration Successful!**

Thank you for registering.